

Maximally Collective Stable Turing Patterns in Binary-Complex Mass-Action Networks

Exact Diffusion Design and Exponent-Optimal Heterogeneity Trade-Offs

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Abstract

Multicomponent Turing theories often seek a lower-dimensional unstable subsystem responsible for patterning. We construct, for every $n \geq 4$, an n -species classical mass-action topology whose source and product complexes contain at most two molecules and whose instability is maximally collective. Uniformly over every positive-equilibrium realization of the topology, every principal subsystem below order $n - 1$ is Hurwitz, whereas one order- $(n - 1)$ subsystem has a positive real eigenvalue. We then derive an exact diffusion-design law: for any homogeneously stable realization, stationary destabilization along a positive diagonal diffusion ray occurs if and only if $d_Z > 8h_Z \sum_{j=2}^{n-2} d_j/h_j$. This yields sharp fixed-equilibrium contrast infima and the universal trade-off $\chi_D \chi_H > 8(n - 3)$. At unit equilibrium we give a rational primary supercritical Turing realization with $\chi_D = 23(91n - 274)/63$ and locally exponentially attracting positive patterns. A second exact family redistributes heterogeneity between diffusion and equilibrium concentrations; both contrasts are $\Theta(\sqrt{n})$, which matches the universal lower bound in exponent. The proof combines a complete positive-realization parametrization, an exhaustive strongly connected block classification, a one-bad-minor matrix theorem, conservation-compatible Lyapunov-Schmidt reduction, and exact symbolic sign certificates. The networks are synthetic, and the bifurcation and stability conclusions are local for each fixed dimension.

1 Introduction

Turing’s diffusion-driven instability mechanism shows how a spatially homogeneous equilibrium can be stable in a well-mixed system and unstable to a nonconstant spatial mode after species-dependent diffusion is introduced [15]. For systems with many components, a central structural question is whether the instability can always be attributed to a smaller unstable subsystem. General matrix theories allow responsible principal subsystems of orders up to $n - 1$ and distinguish stationary from wave instabilities [13, 1, 18, 17]. Reaction-network approaches instead constrain the Jacobian by stoichiometry, positive steady fluxes, and mass-action derivatives [10, 4, 14, 16].

A large unstable block in one numerical Jacobian does not settle the reaction-topology question. Reparameterizing the same network can change its Jacobian and can expose a smaller unstable subsystem. Our first result closes this quantifier exactly: for one explicit indexed topology, every positive rate/equilibrium realization has all principal blocks below order $n - 1$ Hurwitz. Thus the upper endpoint $n - 1$ in principal-subsystem localization cannot be lowered even within a highly restricted classical mass-action class.

The second question is quantitative. Differential diffusion is often presented as an empirical tuning parameter, whereas the present topology admits a complete exact design law. A single signed order- $(n - 1)$ minor is negative, and the remaining omission minors are positive or zero. This “one-bad-minor” structure reduces stationary diffusion design to one strict linear inequality and yields sharp heterogeneity

lower bounds. We further show that the same topology supports primary supercritical bifurcations and locally exponentially stable patterns with substantially reduced diffusion contrast, as well as a stable diffusion–equilibrium trade-off whose two contrasts grow only as the square root of dimension.

Linear Turing instability alone does not guarantee a persistent nonlinear pattern [8]. We therefore prove the primary crossing, the conservation-compatible branch, the exact cubic sign, exchange of stability, and local semilinear stability separately. Related weakly nonlinear analyses for minimal two-species schemes demonstrate the importance of this additional layer [19, 20].

Theorem suite. For every $n = m + 1 \geq 4$, one indexed binary-complex topology has:

1. a complete two-flux-parameter family of all positive-equilibrium Jacobians;
2. topology-wide maximum all-spectrum localization order $n - 1$;
3. an exact necessary-and-sufficient stationary diffusion criterion and a unique positive threshold on each eligible diffusion ray;
4. the sharp fixed-equilibrium diffusion-contrast infimum, including $8(m - 2)$ at the unit equilibrium;
5. a rational unit-equilibrium supercritical stable-pattern realization;
6. a one-parameter stable family with $\chi_D, \chi_H = \Theta(\sqrt{m})$, matching the universal minimax lower bound in exponent.

The topology was deliberately designed so that its positive-flux image and induced strongly connected blocks are explicit; no black-box real-algebraic elimination is used.

Scope. The network has one *semipositive* conservation law: the conserved functional does not contain X_1 . Accordingly we call the system stoichiometric-codimension one, not conservative or mass conserving. The construction is synthetic and not proposed as a natural biochemical mechanism. All species diffuse with positive diagonal coefficients. The pattern, stability, and robustness statements are local for each fixed m ; no arbitrary-data global existence, global attraction, dimension-uniform basin, cross-diffusion, or immobile-species theorem is claimed.

2 Reaction family and complete realization space

An indexed classical mass-action network has reactions $y_r \rightarrow y'_r$, source matrix $Y = [y_1 \cdots y_R]$, and stoichiometric matrix $\Gamma = [y'_1 - y_1 \cdots y'_R - y_R]$. We call it *binary-complex* when

$$|y_r|_1 \leq 2, \quad |y'_r|_1 \leq 2 \tag{1}$$

for every reaction. This condition restricts both source and product molecularity and is stronger than conventions that bound only reactant order.

At a positive equilibrium x^* define

$$v_r = k_r (x^*)^{y_r}, \quad h_i = (x_i^*)^{-1}, \quad H = \text{diag}(h). \tag{2}$$

Then $v > 0$, $\Gamma v = 0$, and direct differentiation gives

$$J = \Gamma \text{diag}(v) Y^T H. \tag{3}$$

Conversely, every $v > 0$ in $\ker \Gamma$ and every $H \succ 0$ are physically realized by $x_i^* = h_i^{-1}$ and $k_r = v_r / (x^*)^{y_r}$.

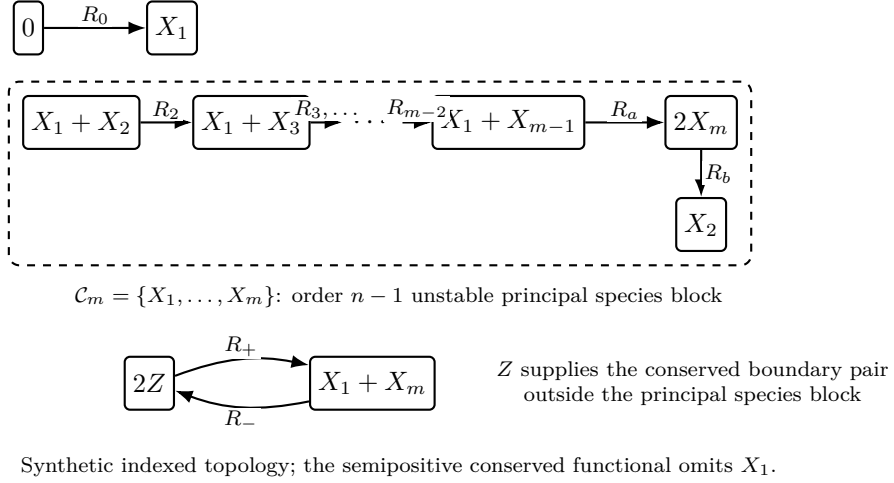


Figure 1: The synthetic topology $\hat{N}_m$. The shaded principal species set $\mathcal{C}_m = \{X_1, \dots, X_m\}$ is the unique order- $(n - 1)$ block with negative signed determinant. The separate Z boundary pair controls the exact diffusion law.

For $m \geq 3$, let $\hat{N}_m$ have species $X_1, \dots, X_m, Z$ and reactions



The chain range is empty at $m = 3$. Writing $n = m + 1$, the family has n species and $n + 1$ indexed reactions and satisfies (1).

Proposition 2.1 (Semipositive conservation and flux cone). *Let*

$$c = (0, \underbrace{4, \dots, 4}_{X_2, \dots, X_{m-1}}, 2, 1)^T. \quad (10)$$

Then

$$c^T \Gamma_m = 0, \quad \text{rank } \Gamma_m = m, \quad \text{im } \Gamma_m = c^\perp, \quad (11)$$

and

$$\ker \Gamma_m = \text{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}. \quad (12)$$

Consequently

$$\{v > 0 : \Gamma_m v = 0\} = \{(a\mathbf{1}_m, b, b) : a, b > 0\}. \quad (13)$$

The functional $c^T x$ is semipositive: it does not bound X_1 .

Proof. Each reaction vector is orthogonal to c . The X_i balance equations along the long circuit successively equate its m reaction fluxes, and the Z balance equates the final reversible pair. The two displayed kernel vectors are independent. Deleting one reaction from each circuit leaves an explicit $m \times m$ triangular stoichiometric minor with nonzero diagonal, so the rank is m . The image and positive-cone statements follow. $\square$

At unit equilibrium and flux $(a\mathbf{1}_m, b, b)$, the vector field is

$$\begin{aligned}
\dot{x}_1 &= a - ax_1x_{m-1} + bz^2 - bx_1x_m, \\
\dot{x}_2 &= -ax_1x_2 + ax_m^2, \\
\dot{x}_i &= ax_1x_{i-1} - ax_1x_i, \\
\dot{x}_m &= 2ax_1x_{m-1} - 2ax_m^2 + bz^2 - bx_1x_m, \\
\dot{z} &= -2bz^2 + 2bx_1x_m.
\end{aligned} \tag{14}$$

The nonzero entries of its Jacobian factor $A_m(a, b)$ are

$$\begin{aligned}
A_{11} &= -(a+b), & A_{1,m-1} &= -a, & A_{1m} &= -b, & A_{1Z} &= 2b, \\
A_{21} &= -a, & A_{22} &= -a, & A_{2m} &= 2a, \\
A_{i,i-1} &= a, & A_{ii} &= -a, & & & & 3 \leq i \leq m-1, \\
A_{m1} &= 2a-b, & A_{m,m-1} &= 2a, & A_{mm} &= -(4a+b), & A_{mZ} &= 2b, \\
A_{Z1} &= 2b, & A_{Zm} &= 2b, & A_{ZZ} &= -4b.
\end{aligned} \tag{15}$$

Proposition 2.2 (Complete realization family). *Every positive-equilibrium Jacobian of $\hat{N}_m$ is exactly*

$$J_m(a, b, H) = A_m(a, b)H, \quad a, b > 0, \quad H \succ 0 \text{ diagonal.} \tag{16}$$

Conversely every triple in (16) is realized by positive rates and a positive equilibrium.

Proof. The forward direction is Eqs. (3) and (13). For the converse, choose $x_i^* = h_i^{-1}$, flux $(a\mathbf{1}_m, b, b)$, and $k_r = v_r/(x^*)^{y_r} > 0$. $\square$

3 Maximum topology-wide all-spectrum localization

For a real square matrix M , write $\alpha(M) = \max\{\Re \lambda : \lambda \in \sigma(M)\}$.

Lemma 3.1 (Exhaustive strongly connected blocks). *For every proper principal species set of cardinality less than m , each nontrivial strongly connected diagonal block of the induced Jacobian graph is one of:*

1. *the long cycle on $X_1, \dots, X_{m-1}$;*
2. *the long cycle on $X_2, \dots, X_m$;*
3. *a principal block of the boundary triad $\{X_1, X_m, Z\}$;*
4. *a negative singleton.*

All remaining induced matrices are block triangular combinations of these. On the hypersurface $b = 2a$, the edge $X_1 \rightarrow X_m$ disappears and only refines this decomposition.

Proof. Interior chain edges point forward. A return path through the lower boundary requires either the complete $X_1, \dots, X_{m-1}$ circuit or the complete $X_2, \dots, X_m$ circuit. Once one long cycle is present, adding a further boundary vertex requires at least m species. Without a complete long cycle, only the three boundary vertices can form a nontrivial strongly connected component. Every other retained vertex has only its negative diagonal inside its component. Removing an edge at $b = 2a$ cannot merge components. $\square$

Theorem 3.2 (Maximal all-spectrum localization). *For every $m \geq 3$, $a, b > 0$, positive diagonal H , and principal set I with $|I| < m$,*

$$\alpha(J_m(a, b, H)_{I, I}) < 0. \quad (17)$$

For $\mathcal{C}_m = \{X_1, \dots, X_m\}$, the block $J_{\mathcal{C}_m, \mathcal{C}_m}$ has a positive real eigenvalue. Thus every positive realization of the topology satisfies

$$\min\{|I| : \alpha(J_{I, I}) > 0\} = m = n - 1. \quad (18)$$

No smaller principal subsystem supports either stationary or oscillatory instability.

Proof. For the first long cycle, a closed-right-half-plane root would force

$$\prod_{i=1}^{m-1} |\lambda + \delta_i h_i| = a^{m-1} \prod_{i=1}^{m-1} h_i, \quad \delta_1 = a + b, \quad \delta_i = a \ (i \geq 2),$$

but the left side is at least $(a + b)a^{m-2} \prod h_i > a^{m-1} \prod h_i$. For the second long cycle, the corresponding diagonal product is $(4a + b)a^{m-2} \prod h_i$, strictly larger than the cycle product $4a^{m-1} \prod h_i$.

The full boundary triad has characteristic polynomial $\lambda^3 + c_1 \lambda^2 + c_2 \lambda + c_3$, where

$$\begin{aligned} c_1 &= ah_1 + 4ah_m + bh_1 + bh_m + 4bh_Z, \\ c_2 &= a(4ah_1h_m + 7bh_1h_m + 4bh_1h_Z + 16bh_mh_Z), \\ c_3 &= 16a^2bh_1h_mh_Z. \end{aligned} \quad (19)$$

All coefficients are positive, and $c_1c_2 - c_3$ is a times fourteen strictly positive monomials (Supplement, Section S2). The cubic Routh–Hurwitz criterion and direct one- and two-dimensional checks establish Hurwitz stability of every triad principal block. The result for all smaller sets follows from Theorem 3.1 and Frobenius block triangularization.

Finally, sparse chain expansion gives

$$(-1)^m \det J_{\mathcal{C}_m, \mathcal{C}_m} = -2a^{m-1}b \prod_{i=1}^m h_i < 0. \quad (20)$$

Its characteristic polynomial is negative at zero and positive for large positive λ , so the block has a positive real eigenvalue. $\square$

4 A one-bad-minor diffusion theorem

For $I \subseteq [n]$, put $a_I = (-1)^{|I|} \det J_{I, I}$ and use the convention $a_\emptyset = 1$.

Theorem 4.1 (One-bad-minor diffusion ray). *Let $J \in \mathbb{R}^{n \times n}$ have one semisimple zero eigenvalue and otherwise stable spectrum. Assume every principal block of order at most $n - 2$ is Hurwitz and, among the signed order- $(n - 1)$ principal minors, at most one is negative. For $D = \text{diag}(d_1, \dots, d_n) \succ 0$, define $p_D(s) = \det(sD - J)$. Then*

$$p_D(s) = s\{\beta_1(D) + \beta_2(D)s + \dots + \beta_n(D)s^{n-1}\}, \quad (21)$$

where

$$\beta_k(D) = \sum_{|I|=n-k} a_I \prod_{j \notin I} d_j, \quad \beta_k(D) > 0 \quad (k \geq 2). \quad (22)$$

A nonzero stationary threshold exists on the ray sD if and only if $\beta_1(D) < 0$. In that case there is a unique simple $s_ > 0$, and*

$$J - sD \text{ has a positive real eigenvalue} \iff 0 < s < s_*. \quad (23)$$

For $s > s_$ no positive real eigenvalue exists. The theorem does not exclude a nonreal unstable pair outside this interval.*

Proof. The principal-minor expansion of $\det(sD - J)$ gives Eqs. (21) and (22); the constant term vanishes because $\det J = 0$. Every a_I of order at most $n - 2$ is positive, hence $\beta_k > 0$ for $k \geq 2$. The bracket in (21) is strictly increasing for $s > 0$, proving existence, uniqueness, signs, and scalar simplicity of s_* . Since $p_D(s) = \det(D) \det(sI - D^{-1}J)$, scalar simplicity is algebraic simplicity of the zero eigenvalue.

For fixed $s > 0$,

$$\chi_s(\lambda) = \det(\lambda I + sD - J) = \sum_I a_I \prod_{j \notin I} (\lambda + sd_j). \quad (24)$$

All derivative contributions from $|I| \leq n - 2$ are nonnegative, with a strictly positive contribution. The total derivative contribution from $|I| = n - 1$ is $\sum_{|I|=n-1} a_I = \beta_1(\mathbf{1}) > 0$: equal diagonal damping shifts every stable nonzero eigenvalue and the simple zero strictly left, so the coefficient of s in $\det(sI - J)$ is positive. Thus $\chi'_s(\lambda) > 0$ for $\lambda \geq 0$. Since $\chi_s(0) = p_D(s)$ and $\chi_s(\lambda) \rightarrow \infty$, the positive-real-root equivalence (23) follows. $\square$

5 Exact diffusion design and sharp linear contrast

The all-spectrum theorem supplies the lower-order hypotheses of Theorem 4.1. The remaining information is the complete order- m omission table.

Proposition 5.1 (Order- $(n - 1)$ omission minors). *For $J = A_m(a, b)H$,*

$$(-1)^m \det J_{\widehat{Z}, \widehat{Z}} = -2a^{m-1}b \prod_{i=1}^m h_i, \quad (25)$$

$$(-1)^m \det J_{\widehat{X_j}, \widehat{X_j}} = 16a^{m-1}bh_Z \prod_{\substack{1 \leq i \leq m \\ i \neq j}} h_i, \quad 2 \leq j \leq m - 1, \quad (26)$$

$$\det J_{\widehat{X_1}, \widehat{X_1}} = \det J_{\widehat{X_m}, \widehat{X_m}} = 0. \quad (27)$$

Consequently

$$\beta_1(D) = 2a^{m-1}b \left(\prod_{i=1}^m h_i \right) \left[8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} - d_Z \right]. \quad (28)$$

Proof. Omitting Z yields the chain expansion (20). Omitting an interior X_j breaks the chain into negative singleton factors and the remaining boundary triad contributes signed determinant $16a^2b$. Omitting X_1 leaves the restricted conservation vector as a left nullvector; omitting X_m leaves the restriction of the full right kernel vector, whose X_m coordinate is zero. Multiplication by the omitted diffusion coefficient and summation gives (28). The coefficient 8 is the exact ratio $16/2$ between the positive interior-omission weight and the magnitude of the unique negative Z -omission weight. $\square$

Theorem 5.2 (Exact stationary diffusion law). *Assume $J_m(a, b, H)$ is stable on its stoichiometric subspace. A nonzero stationary crossing occurs on the positive ray sD if and only if*

$$d_Z > 8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j}. \quad (29)$$

When (29) holds, the threshold $s_*(H, D)$ is unique and simple, and the exact positive-real-eigenvalue interval is $0 < s < s_*$. Scaling $D \mapsto \gamma D$ sends $s_* \mapsto s_*/\gamma$.

Proof. Combine Theorems 4.1 and 5.1. A second derivation uses the left vector c and the right zero vector

$$r_0 = H^{-1}(2, -2, \dots, -2, 0, 1)^T.$$

Then

$$\lambda'_0(0) = -\frac{c^T D r_0}{c^T r_0}, \quad c^T D r_0 = \frac{d_Z}{h_Z} - 8 \sum_{j=2}^{m-1} \frac{d_j}{h_j},$$

while

$$c^T r_0 = h_Z^{-1} - 8 \sum_{j=2}^{m-1} h_j^{-1} < 0.$$

The last inequality follows from the positive equal-damping coefficient $\beta_1(\mathbf{1}) > 0$. Hence the conservation eigenvalue initially moves right exactly under (29). $\square$

For a Neumann interval $(0, L)$, mode k has $s = q_k^2 = (k\pi/L)^2$. Therefore every mode with $q_k^2 < s_*$ has a positive real eigenvalue, and no mode with $q_k^2 > s_*$ has one. A separate spectral certificate is required to exclude wave instability for a chosen diffusion profile.

Theorem 5.3 (Sharp contrast and heterogeneity trade-off). *Define*

$$T(H) = 8h_Z \sum_{j=2}^{m-1} \frac{1}{h_j}, \quad \chi_D = \frac{d_{\max}}{d_{\min}}, \quad \chi_H = \frac{h_{\max}}{h_{\min}} = \frac{x_{\max}^*}{x_{\min}^*}. \quad (30)$$

Homogeneous stability implies $T(H) > 1$, and

$$\inf\{\chi_D : D \succ 0 \text{ satisfies (29)}\} = T(H). \quad (31)$$

The infimum is not attained. At $H = I$,

$$\inf \chi_D = 8(m-2). \quad (32)$$

Every stationary crossing satisfies the universal strict trade-off

$$\boxed{\chi_D \chi_H > 8(m-2)}. \quad (33)$$

The product lower bound is sharp as an infimum over stable realizations.

Proof. If (29) holds, then $d_Z > d_{\min} T(H)$ and $d_{\max} \geq d_Z$, so $\chi_D > T(H)$. Conversely set all X -species diffusivities to one and $d_Z = T(H) + \varepsilon$. This proves (31) and its strict nonattainment. The unit formula follows immediately. Since $h_Z/h_j \geq 1/\chi_H$, one has $T(H) \geq 8(m-2)/\chi_H$, yielding (33). Taking $H = I$ and the approaching diffusion sequence shows sharpness of the product infimum. Equal diffusion cannot destabilize because $J - sdI$ shifts the stable nonzero spectrum and the conservation zero strictly left. $\square$

6 A supercritical stable unit-equilibrium design

Set $a = b = 1$, $H = I$, $x^* = \mathbf{1}$, and

$$K_i = 91m - 181 - i. \quad (34)$$

Define the critical vector

$$r_1 = 1, \quad r_i = -\frac{K_i}{63(m-2)} \quad (2 \leq i \leq m-1), \quad r_m = -\frac{2}{9}, \quad r_Z = \frac{5}{14}, \quad (35)$$

and diffusion profile

$$d_1 = \frac{23}{63}, \quad d_i = \frac{1}{K_i} \quad (2 \leq i \leq m-1), \quad d_m = \frac{1}{7}, \quad d_Z = \frac{16}{45}. \quad (36)$$

All entries are positive. Here $d_{\min} = d_2 = (91m - 183)^{-1}$ and $d_{\max} = d_1 = 23/63$, so

$$\chi_D^{\text{unit}}(m) = \frac{23(91m - 183)}{63}. \quad (37)$$

Theorem 6.1 (Primary supercritical stable design). *On $(0, \pi)$ with Neumann conditions, consider*

$$\partial_t x = f_m(x) + (1 - \mu)D_m \partial_{\xi\xi} x, \quad (38)$$

with (36). For every $m \geq 3$:

1. *the well-mixed Jacobian, equivalently the homogeneous $k = 0$ mode restricted to $c^\perp$, is stable;*
2. *at $\mu = 0$ the first nonzero mode has one algebraically simple zero with right vector (35); every other first-mode eigenvalue and every higher mode is stable, and no complex pair is critical;*
3. *the crossing is transverse;*
4. *a conservation-compatible Lyapunov–Schmidt reduction has*

$$\dot{A} = \eta_m \mu A + c_m A^3 + O(|A|^5 + |A|^3|\mu| + |A||\mu|^2), \quad \eta_m > 0, \quad c_m < 0; \quad (39)$$

5. *for sufficiently small $\mu > 0$, the two positive nonconstant branches are locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space.*

Consequently

$$8(m - 2) \leq \chi_{\text{stable}}(m) \leq \frac{23(91m - 183)}{63}. \quad (40)$$

Proof architecture. The homogeneous characteristic determinant is

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \quad \lambda q_m = (1 + \lambda)^{m-3} P(\lambda) - R(\lambda), \quad (41)$$

where $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. The exact difference $|1 + \lambda|^2 |P|^2 - |R|^2$ has 35 nonnegative monomials in $\Re \lambda$ and $(\Im \lambda)^2$, with equality only at the origin.

For damping tD_m , sparse chain elimination gives

$$\det(\lambda I - A_m + tD_m) = Q_m F - G, \quad (42)$$

where $Q_m = \prod_{i=2}^{m-1} (\lambda + 1 + t/K_i)$ and the fixed boundary factors are

$$g_1 = \lambda + 2 + \frac{23}{63}t, \quad g_m = \lambda + 5 + \frac{1}{7}t, \quad g_Z = \lambda + 4 + \frac{16}{45}t,$$

$$F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z, \quad G = g_Z(4g_1 + g_m) - 36.$$

At onset $\prod_{i=2}^{m-1} (1 + 1/K_i) = 91/90$. Retaining one exact chain factor yields a 77-term nonnegative modulus certificate, strict except at $(\lambda, t) = (0, 1)$. This proves the complete primary mode statement. An exact left vector is

$$\ell_1 = -\frac{266}{815}, \quad \ell_i = \frac{78260(m-2)}{163(91m-180-i)}, \quad \ell_m = \frac{18368}{7335}, \quad \ell_Z = 1, \quad (43)$$

with $2 \leq i \leq m-1$. Exact harmonic-sum formulas give $\ell^T r < 0$ and $\ell^T D_m r < 0$, hence $\eta_m = (\ell^T D_m r)/(\ell^T r) > 0$.

The reaction field is quadratic: $f_m(\mathbf{1} + u) = A_m u + \frac{1}{2} B_m(u, u)$, with $c^T B_m = 0$. Since $\cos^2 \xi = (1 + \cos 2\xi)/2$, the stable corrections solve

$$A_m w_0 = -\frac{1}{4} B_m(r, r), \quad c^T w_0 = 0, \quad (A_m - 4D_m) w_2 = -\frac{1}{4} B_m(r, r). \quad (44)$$

The interior w_2 recurrence telescopes through four consecutive K_i factors. Fourier projection gives

$$c_m = \frac{\ell^T \{B_m(r, w_0) + \frac{1}{2}B_m(r, w_2)\}}{\ell^T r}. \quad (45)$$

Its numerator reduces to $R_m + C_m \mathcal{H}_m$ with $\mathcal{H}_m = \sum_{j=1}^{m-2} K_j^{-1}$. The exact shifted-polynomial certificate proves the numerator positive and the denominator negative for all $m \geq 3$, hence $c_m < 0$.

Crandall–Rabinowitz gives branches $A_{\pm}(\mu) = \pm\sqrt{-\eta_m\mu/c_m} + O(\mu^{3/2})$. The center eigenvalue is $-2\eta_m\mu + O(\mu^2) < 0$; the complementary gap persists. On the real fixed-mass L^2 space, positive Neumann diffusion with domain H_N^2 is sectorial, while in one dimension the quadratic Nemytskii map is smooth from H^1 to L^2 . The principle of linearized stability then gives local exponential asymptotic stability in H^1 [3, 6, 5, 7]. The complete coefficient tables and algebra appear in the Supplement. $\square$

7 A stable diffusion–equilibrium trade-off family

Put $r = m - 2$ and retain the effective diffusion matrix Δ_m from (36). Define

$$L_0(r) = \frac{1}{\sqrt{3r}}, \quad L_1(r) = \frac{90r}{90r + 1}. \quad (46)$$

For $L \in [L_0, L_1]$, set

$$h_1 = h_m = h_Z = 1, \quad h_i = \frac{K_i}{LK_{i-1}} \quad (2 \leq i \leq m-1), \quad H_m(L) = \text{diag}(h), \quad (47)$$

and choose

$$D_m^{\text{phys}}(L) = H_m(L)\Delta_m, \quad x^* = H_m(L)^{-1}\mathbf{1}. \quad (48)$$

Positive rates realizing unit reaction flux are given by $k_r = (x^*)^{-y_r}$.

Theorem 7.1 (Stable exponent-optimal trade-off). *For every $m \geq 3$ and $L \in [L_0(r), L_1(r)]$, the physical realization (47)–(48) has a primary simple transverse first-mode stationary bifurcation, a negative cubic coefficient, and two nearby positive locally exponentially asymptotically stable patterned equilibria. Its physical contrasts are*

$$\chi_D(L) = \frac{23}{63} 91rL, \quad (49)$$

$$\chi_H(L) = \frac{91r - 1}{91rL}, \quad (50)$$

$$\chi_D(L)\chi_H(L) = \frac{23(91r - 1)}{63}. \quad (51)$$

At $L = L_0$,

$$\chi_D = \frac{2093}{63\sqrt{3}}\sqrt{r}, \quad \chi_H = \sqrt{3r} \frac{91r - 1}{91r}. \quad (52)$$

Thus both contrasts are $\Theta(\sqrt{m})$. Since every stationary crossing obeys

$$\max(\chi_D, \chi_H) > \sqrt{8(m-2)}, \quad (53)$$

the simultaneous exponent $1/2$ is globally optimal. No constant-optimality or complete global Pareto-frontier claim is made.

Proof architecture. In normalized concentrations $z = H_m(L)x$, the PDE is

$$z_t = H_m(L)\{f_m(z) + \Delta_m z_{\xi\xi}\}. \quad (54)$$

The stationary bracket $f_m(z) + \Delta_m z_{\xi\xi}$ and its critical right vector are unchanged. The dynamic left vector is $q_m(L) = H_m(L)^{-1}\ell_m$, and the physical fixed-mass vector becomes $H_m(L)^{-1}c$.

For $2 \leq i \leq m-1$, chain elimination gives

$$g_i = \frac{\lambda}{h_i} + 1 + \frac{t}{K_i} = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

If $t \geq 1$ and $\lambda = x + iy$ with $x \geq 0$, then

$$|Q_m|^2 \geq \left(\frac{91}{90}\right)^2 |1 + L\lambda|^{2r} \geq \left(\frac{91}{90}\right)^2 \left(1 + 2rLx + \frac{1}{3}y^2\right),$$

because $rL^2 \geq 1/3$. The corresponding boundary difference has 84 nonnegative monomials and equality only at $(\lambda, t) = (0, 1)$. A separate 34-term certificate proves homogeneous relative stability. Hence the primary mode and all higher-mode conclusions hold throughout the certified interval.

The second harmonic is unchanged by the invertible row factor. The homogeneous correction changes by a kernel gauge:

$$w_0(L) = w_0^{\text{ref}} + \tau_m(L)\rho, \quad \tau_m(L) = -\frac{(H_m(L)^{-1}c)^T w_0^{\text{ref}}}{(H_m(L)^{-1}c)^T \rho}. \quad (55)$$

The exact cubic numerator is

$$N_m(L) = N_m^{\text{ref}} + \tau_m(L)S_m. \quad (56)$$

The symbolic bounds $N_m^{\text{ref}} > 1/100$, $-1/10 < S_m < 0$, and $\tau_m(L) < 1/20$ imply $N_m(L) > 1/200$. Meanwhile

$$\ell_m^T H_m(L)^{-1} r_m = -\frac{485873}{924210} - \frac{11180}{1467} L(m-2) < 0.$$

Therefore $\eta_m(L) > 0$ and $c_m(L) < 0$. The branch and stability proof is the same sectorial exchange-of-stability argument as in Theorem 6.1. The contrast formulas follow by comparing the explicit entries of $H_m(L)$ and $D_m^{\text{phys}}(L)$ on the interval (46). Finally (53) is the geometric-mean consequence of (33). $\square$

8 The nonlinear frontier and finite-dimensional transparency

The exact unit-equilibrium stable infimum is open. The completed bounds are

$$8(m-2) \leq \chi_{\text{stable}}(m) \leq \frac{23(91m-183)}{63}. \quad (57)$$

A canonical near-threshold affine path has, at $m = 3$,

$$c_3(\varepsilon) = \frac{6}{1379} + \frac{421985}{11409846}\varepsilon + O(\varepsilon^2), \quad c_3(\varepsilon) > 0 \quad (0 < \varepsilon \leq 10^{-3}). \quad (58)$$

Thus approaching the sharp linear threshold does not automatically preserve supercriticality. This is a boundary example, not a universal nonlinear gap.

Proposition 8.1 (Retuned local robustness). *Fix m and a certified L . Sufficiently small perturbations of positive rate constants, equilibrium coordinates, diffusion ratios, and interval length admit a unique nearby scalar diffusion multiplier at which the primary supercritical bifurcation and local exponential stability persist.*

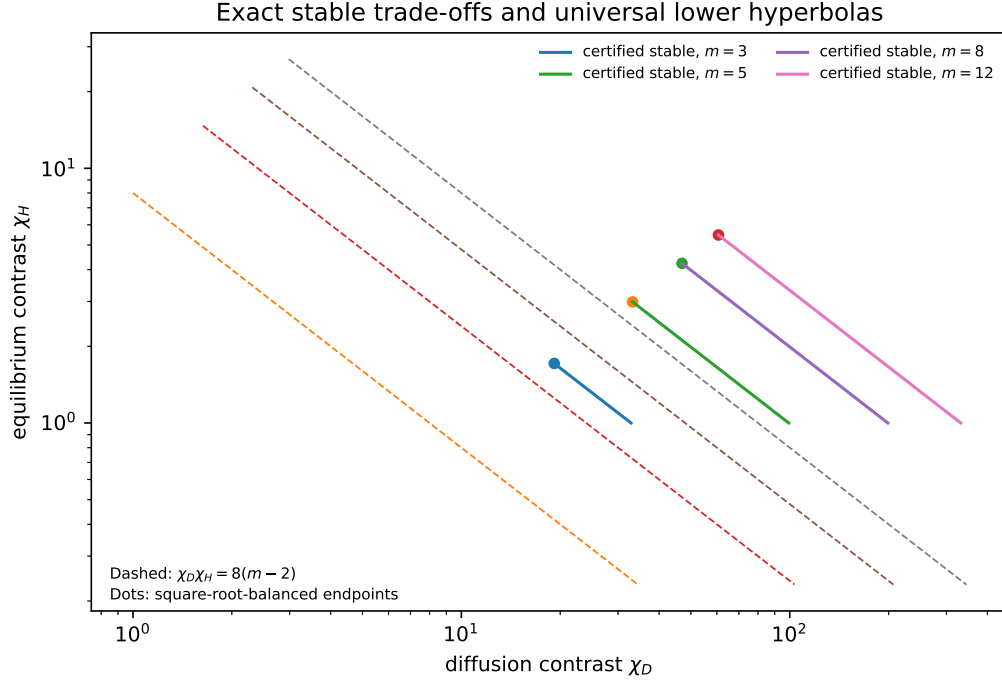


Figure 2: Exact heterogeneity trade-offs for representative dimensions. The dashed hyperbolas are the universal necessary bounds $\chi_D \chi_H = 8(m-2)$; the solid curves are certified stable families with product $23(91(m-2) - 1)/63$. The marked point is the square-root-balanced endpoint. Constants are not claimed optimal.

Proof. Continue the simple critical eigenvalue as a smooth function of the physical parameters and one scalar diffusion multiplier. Its multiplier derivative is the nonzero transversality pairing. The implicit function theorem gives the nearby critical multiplier and equilibrium. Homogeneous and finitely many low-mode gaps persist by analytic perturbation. In a common neighborhood, a positive diffusion lower bound and a reaction-Jacobian upper bound control all high modes. The Lyapunov–Schmidt coefficients and branch spectral projection depend smoothly on parameters, so the negative cubic sign and complementary branch gap persist. One scalar multiplier is retuned; arbitrary nearby parameter points are not asserted to be critical. $\square$

Open problems.

1. Determine whether $\chi_{\text{stable}}(m) = 8(m-2)$ as an infimum at unit equilibrium.
2. Determine whether supercritical stable pattern formation has a strict nonlinear contrast gap above the linear threshold.
3. Determine the constant-optimal stable diffusion–equilibrium frontier, or equivalently the exact infimum of $\max(\chi_D, \chi_H)$.

9 Numerical illustrations

The symbolic theorem does not use simulation. We integrated cosine–Galerkin systems for $m = 3, 5, 8$ near the exact unit-equilibrium branches, extracted the first-mode amplitude with the adjoint projection, and repeated the calculations under spatial and temporal refinement. Parameters, tolerances, profiles, and open-format data accompany the code.

m	n	χ_D^{unit}	χ_D^{bal}	χ_H^{bal}	product	lower	η_m	c_m	$\sqrt{-\eta_m/c_m}$
3	4	32.86	19.18	1.713	32.86	2.828	0.1054	-0.0618	1.306
4	5	66.08	27.13	2.436	66.08	4	0.05706	-0.0333	1.309
5	6	99.3	33.22	2.989	99.3	4.899	0.03911	-0.0228	1.31
6	7	132.5	38.36	3.455	132.5	5.657	0.02976	-0.01733	1.31
7	8	165.7	42.89	3.864	165.7	6.325	0.02401	-0.01398	1.31
8	9	199	46.98	4.235	199	6.928	0.02013	-0.01172	1.311
9	10	232.2	50.75	4.575	232.2	7.483	0.01732	-0.01008	1.311
10	11	265.4	54.25	4.892	265.4	8	0.0152	-0.008849	1.311

Table 1: Exact-design diagnostics. χ_D^{unit} is (37); the balanced values use $L = L_0$ in (52). The normal-form values are for the unit-equilibrium stable profile. The table illustrates exact formulas but is not used to prove any all-dimensional statement.

10 Relation to prior work and limitations

The possibility of order- $(n - 1)$ responsible subsystems for general fixed matrices is established in multicomponent Turing theory [13, 1]; that fixed-matrix possibility is not claimed as new. The present distinction is the conjunction of one indexed binary-complex classical mass-action topology, every positive rate and equilibrium realization, all-spectrum stability of every smaller principal block, exact diffusion design, and stable nonlinear patterned branches in every dimension.

Mincheva and Craciun’s principal projected-network injectivity condition is a different reaction-network notion and is not identified with principal-block Hurwitz localization [10]. Interaction-topology and atlas studies identify broad structural features and finite collections of Turing-capable networks [9, 4, 14]; the present result instead gives one exact infinite topology family with a universal over-realizations theorem. Parameter-rich unstable cores allow derivative freedoms absent from classical mass action [16]. Exact fixed-three-species criteria address a different, fixed-Jacobian scope [12]. Biochemical screens study interpretable reaction architectures rather than the synthetic extremal design here [11]. Monomial-steady-state results give sufficient spatial instability conditions for a restricted mass-action class [2].

The construction has one semipositive conservation law and an inflow; X_1 is not bounded by the conserved functional. The local stability theorem does not require an arbitrary-data global boundedness theorem. All diffusivities are positive and diagonal. We do not treat immobile species, cross-diffusion, projected-injectivity sharpness, weak reversibility, biochemical realization, far-from-onset dynamics, global attraction, or dimension-uniform robustness. The exact stable contrast constants remain open.

11 Discussion

The family is designed around two coupled positive-flux circuits. Their explicit two-parameter Jacobian image makes the topology-wide quantifier tractable. The long chain forces every unstable principal subsystem to contain almost all species, while the boundary triad remains Hurwitz under arbitrary positive equilibrium scaling. The unique negative order- $(n - 1)$ omission minor is the one omitting Z , which collapses stationary diffusion design to one coefficient. Finally, equilibrium scaling redistributes the necessary heterogeneity between concentrations and physical diffusion without changing the effective critical bracket or losing the negative cubic sign.

The resulting conclusion is both structural and quantitative. The largest proper unstable-subsystem order is unavoidable within this mass-action class; the exact diffusion cost is computable; stable positive patterns occur; and diffusion and equilibrium contrasts can be balanced at the globally optimal square-root exponent. The remaining constant gap is explicit rather than hidden and provides a focused nonlinear design problem.

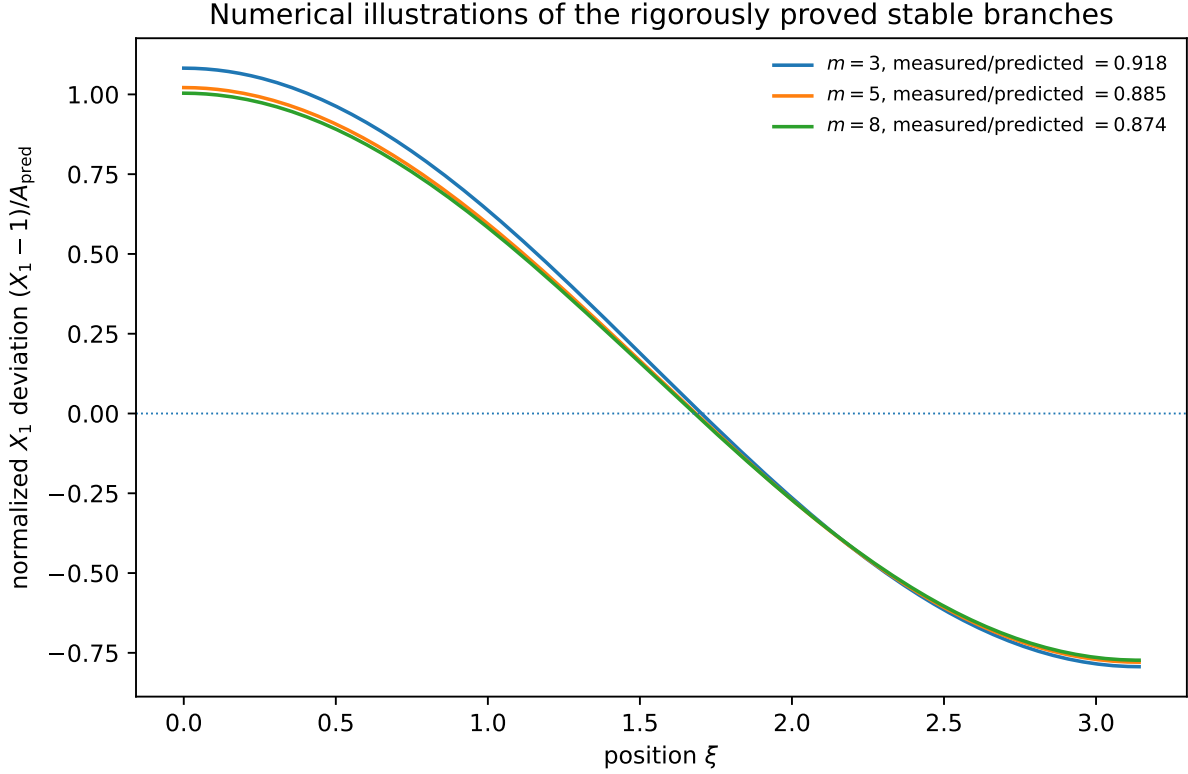


Figure 3: Numerical illustrations of the rigorously proved local patterned branches. The right panel compares the measured adjoint-projected amplitude with $\sqrt{-\eta_m \mu / c_m}$. Numerical illustrations are not used in any proof.

Data and code availability. Exact certificates, independent verifiers, mutation tests, numerical data, figure scripts, and a one-command replay accompany the manuscript.

AI-assisted research disclosure. AI systems assisted with exploratory theorem discovery, symbolic derivation, code generation, adversarial testing, proof organization, and prose editing. AI systems are not authors. The human author assumes responsibility for the manuscript. Central claims are supported by explicit proofs and independent exact implementations; neither numerical agreement nor AI confidence is treated as proof.

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